

EAGLE – WP2.3: Open-source software comparison

Purpose of the comparison. The software must ultimately support the WP2.3 chain: *Spanish network data* → *electrical network model* → *power-system analysis* → *line/voltage/grid constraints* → *nodal hosting-capacity assessment* → *spatial indicators for SiteOptimizer*. The project proposal specifically targets HV and, where possible, MV network data, line saturation, regional flexibility requirements and nodal grid hosting capacity.

Colour code: **YES** capability is well aligned with the EAGLE requirement; **PARTIAL** capability exists or can be achieved, but is secondary, indirect, requires additional workflow, or is not the tool's main strength; **NO** the tool does not provide that function by itself.

Requirement	PyPSA	pandapow	OpenDSS	MATPOW	GridCal	Comment for EAGLE
Open-source	YES	YES	YES	YES	YES	All five are open-source tools and therefore compatible with the project's open-source requirement.
Python / programmatic workflow	YES	YES	PARTIAL	PARTIAL	YES	Python is especially convenient for EAGLE because data acquisition, preprocessing, modelling and SiteOptimizer are expected to be highly programmatic.
Create/import buses, lines, transformers and loads	YES	YES	YES	YES	YES	Core requirement for reconstructing the Spanish electrical network.
HV transmission modelling	YES	YES	PARTIAL	YES	YES	PyPSA, pandapower, MATPOWER and GridCal are directly relevant to transmission-level modelling; OpenDSS is primarily distribution-oriented.
MV / distribution modelling	PARTIAL	YES	YES	PARTIAL	YES	Important if EAGLE extends the initial HV model towards MV.
AC power flow	PARTIAL	YES	YES	YES	YES	A key requirement for checking voltages and branch flows. PyPSA's current core is strongly oriented to linearised network formulations; this should be checked against the exact study methodology.
DC / linear power-flow capability	YES	YES	PARTIAL	YES	YES	Useful for large-scale screening and repeated spatial scenarios.
Optimal Power Flow (OPF)	YES	YES	PARTIAL	YES	YES	Relevant for constrained operation and optimisation-based capacity studies.
Time-series simulation	YES	YES	YES	PARTIAL	YES	Useful because grid constraints can depend on time-varying demand and renewable generation.
Line loading / thermal-limit analysis	YES	YES	YES	YES	YES	Directly linked to T2.3's explicit focus on line saturation / grid constraints.
Voltage / operational constraints	YES	YES	YES	YES	YES	Needed to define feasible connection/capacity conditions, not just line loading.

Contingency / security analysis	YES	YES	PARTIAL	YES	YES	Potentially important later if the hosting-capacity methodology includes security criteria. Exact EAGLE requirement is not yet fixed.
Hosting-capacity assessment	PARTIAL	PARTIAL	YES	PARTIAL	PARTIAL	Hosting capacity is primarily a methodology built on repeated power-flow/OPF/constraint calculations; it is not equivalent to a single built-in button. OpenDSS has particularly strong HC use in the literature.
Large-scale network capability	YES	YES	PARTIAL	YES	YES	EAGLE needs a country/European-scale model rather than a small feeder only.
Optimisation / planning	YES	YES	PARTIAL	YES	YES	Relevant because the final EAGLE framework includes long-term planning and multi-criteria localisation.
Geographic coordinates / spatial network attributes	YES	YES	PARTIAL	PARTIAL	YES	Essential for converting electrical results into spatial indicators for SiteOptimizer.
GIS integration / map visualisation	PARTIAL	YES	PARTIAL	NO	PARTIAL	GIS is not the same as electrical simulation. QGIS/GeoPandas can be used alongside the electrical solver.
Import/open-network-data compatibility	YES	YES	YES	YES	YES	All can be populated from external data; the practical question is how much conversion/preprocessing is required.
Export node/line results for SiteOptimizer	YES	YES	PARTIAL	PARTIAL	YES	The preferred workflow should expose machine-readable node/line results, coordinates and constraint indicators.
Overall role in EAGLE	Strong candidate	Strong candidate	Specialist / secondary	Strong solver	Strong candidate	This is a working screening, not the final software decision.

Interpretation note. The colours are a *working EAGLE-oriented classification*, not a claim that a red/orange tool is generally “bad”. They answer the narrower question: *how directly does this tool satisfy the requirements we currently understand for WP2.3?* Hosting capacity itself is a study methodology and may need to be implemented on top of a power-flow/OPF engine.

Sources checked: PyPSA official documentation and website; pandapower official documentation; MATPOWER documentation; OpenDSS official documentation; the EAGLE proposal supplied in the project; and the Consensus literature screening supplied in this conversation.

EAGLE – WP2.3: Open data sources for the Spanish grid

Purpose of the comparison. The data source should help construct a geographically referenced electrical model of Spain capable of supporting later grid-constraint and hosting-capacity analysis. Based on the project, the most important information includes HV topology (and MV where possible), buses/substations, line connections, voltage levels, geographical location, electrical parameters, line capacity/limits, generation and demand.

Colour code: **YES** useful/valid for the requirement; **PARTIAL** partial, incomplete, indirect or requiring combination with another source; **NO** not provided by the source or not its purpose.

Data requirement	PyPSA-Eur	GridKit	MapYourG / OSM	SciGRID	OpenGrid	REE	ENTSO-E	Comment
Spain coverage	YES	YES	YES	YES	PARTIAL	YES	YES	Spain is within the European/Spanish coverage of the relevant network sources; exact version and scope must be checked.
HV transmission topology	YES	YES	PARTIAL	YES	PARTIAL	YES	YES	PyPSA-Eur directly provides a transmission-level model; GridKit/SciGRID are particularly useful for topology extraction.
MV network detail	NO	PARTIAL	PARTIAL	PARTIAL	PARTIAL	PARTIAL	PARTIAL	PyPSA-Eur explicitly represents AC lines at/above 220 kV; MV requires additional sources/data.
Buses / substations	YES	YES	YES	YES	YES	YES	YES	Core topology information is available, but field completeness differs between sources.
Line connections / topology	YES	YES	YES	YES	PARTIAL	YES	YES	GridKit explicitly converts mapped infrastructure into connected topology.
Geographical coordinates / geometry	YES	YES	YES	YES	YES	YES	YES	Critical for the spatial assessment and later SiteOptimizer GIS layer.
Voltage level	YES	YES	YES	YES	PARTIAL	YES	YES	Voltage information is useful for filtering the network by HV level; OSM-derived sources may have missing/ambiguous tags.
Line length	YES	PARTIAL	YES	YES	PARTIAL	YES	YES	Useful for reconstructing electrical parameters and validating geometry.

Electrical R/X/B parameters	YES	PARTIAL	PARTIAL	YES	NO	PARTIAL	PARTIAL	This is a major discriminator: some topology sources require assumptions or external data for impedance.
Thermal / apparent-power limits	YES	PARTIAL	PARTIAL	PARTIAL	NO	YES	PARTIAL	A critical field for line-loading/capacity studies. Availability and definition must be checked at asset level.
Generation connected to network	YES	PARTIAL	PARTIAL	PARTIAL	PARTIAL	YES	YES	PyPSA-Eur includes conventional plants and renewable availability; REE/ENTSO-E provide complementary system information.
Spatial demand / load	YES	NO	PARTIAL	PARTIAL	PARTIAL	YES	YES	Needed to establish the baseline operating state before adding hypothetical data-centre loads.
Time-series demand	YES	NO	NO	PARTIAL	NO	YES	YES	Important for scenario/time-series studies; topology datasets alone do not provide this.
Time-series renewable generation / availability	YES	NO	NO	PARTIAL	NO	YES	YES	Useful for time-dependent congestion and hosting-capacity scenarios.
Existing access/hosting-capacity information	PARTIAL	NO	NO	NO	NO	YES	YES	REE currently publishes demand and generation access-capacity information by transport-grid node; this is valuable as a benchmark/validation layer.
Directly usable as an electrical model	YES	PARTIAL	NO	PARTIAL	NO	PARTIAL	PARTIAL	PyPSA-Eur is already a model dataset; GIS/topology sources normally require conversion and parameter completion.
Useful GIS/spatial layer	YES	YES	YES	YES	YES	YES	YES	Several sources are complementary rather than alternatives.

Likely EAGLE role	Base model	Topology	GIS comple-ment	Topology + pa-rameters	GIS comple-ment	Spanish reference	European reference	Likely multi-source workflow
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Important data-quality note. The EAGLE proposal explicitly identifies poor quality, data gaps and insufficient granularity as risks for WP2–WP4. Therefore, a source being marked **YES** does not mean that every required field is guaranteed for every Spanish asset; it means that the source is a credible/usable source for that data category and should be investigated further.

Particularly important finding for the next phase: PyPSA-Eur's current transmission representation is at/above 220 kV, while Red Eléctrica reports a Spanish transport network including 400 kV and 220 kV on the mainland and lower transport voltages in non-peninsular systems. Therefore, the Spanish model may need a combination of PyPSA-Eur/other open topology data with Spanish sources if EAGLE requires more detail than the European transmission model provides.

Sources checked: PyPSA-Eur documentation; GridKit/Zenodo and GridKit repository; MapYourGrid; Red Eléctrica public data and network/access-capacity pages; European Commission EIGL documentation describing GridKit and SciGRID; and the EAGLE proposal supplied in the project.